



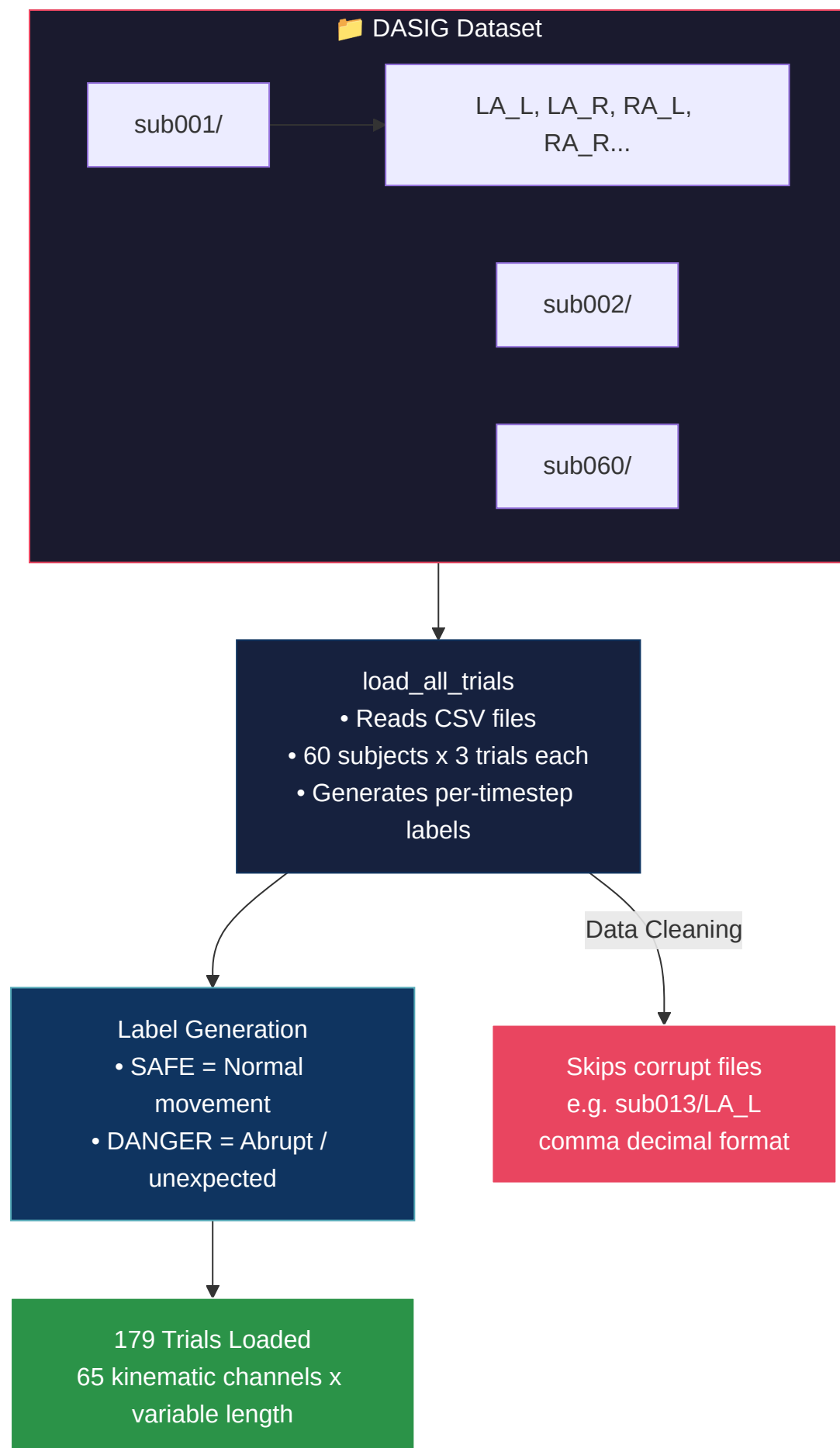
# Cobot Safety-State Detection — Full Pipeline Architecture

Deep Learning Pipeline v4 — Factory-Calibrated (Subject-Dependent)  
Benchmark

## 1. End-to-End Pipeline Overview



## 2. Data Ingestion & Labelling



| Parameter       | Value                                             |
|-----------------|---------------------------------------------------|
| Sensor Channels | <b>65</b> (joint angles, positions, orientations) |
| Sampling Rate   | <b>200 Hz</b>                                     |

| Parameter      | Value |
|----------------|-------|
| Total Subjects | 60    |
| Total Trials   | 179   |

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### 3. Windowing & Normalisation

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179 Trials  
65 channels x variable  
length

- segment\_all\_trials
- Window Size: 0.5s = 100 timesteps
  - Step Size: 0.25s = 50% overlap
  - Label Strategy: all

- Per-Trial Z-Score  
Normalisation
- $\mu$  = mean of trial
  - $\sigma$  = std of trial
  - $x_{\text{norm}} = x - \mu / \sigma$

- Window-Level Label
- $y_{\text{window}} = \max$  of  $y_{\text{timesteps}}$
  - If ANY timestep is DANGER then entire window = DANGER

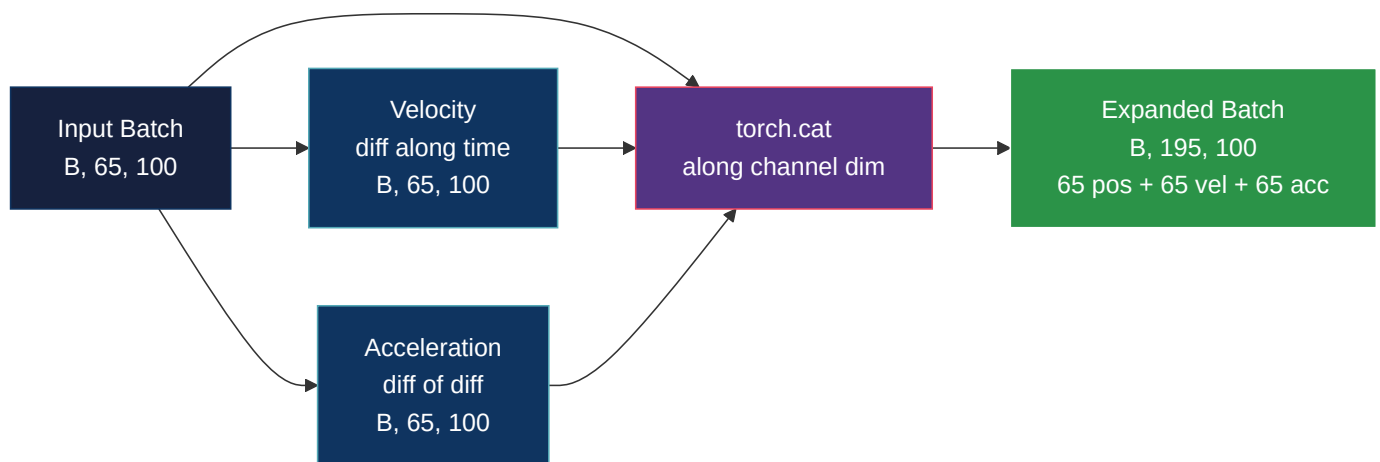
- Final Tensors
- X: 69094 x 65 x 100  
Batch, Channels, SeqLen
  - y: 69094  
Binary: 0=SAFE,  
1=DANGER

Class Distribution

- SAFE: 60559 = 87.6%
- DANGER: 8535 = 12.4%

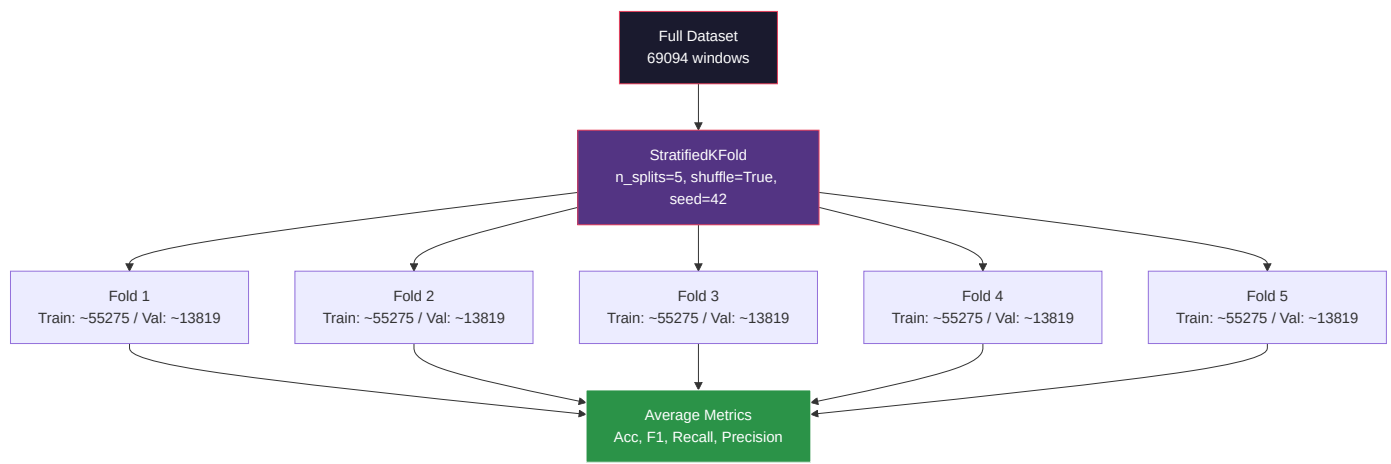
## 4. Dynamic Feature Engineering (On-GPU)

To prevent OOM crashes, velocity and acceleration derivatives are computed **dynamically inside the training loop on the GPU** rather than being pre-computed in RAM.



| Feature Group                 | Channels | Description                 |
|-------------------------------|----------|-----------------------------|
| Position (raw)                | 65       | Original kinematic channels |
| Velocity (1st derivative)     | 65       | Rate of change of position  |
| Acceleration (2nd derivative) | 65       | Jerk detection              |
| Total                         | 195      | Fed into every model        |

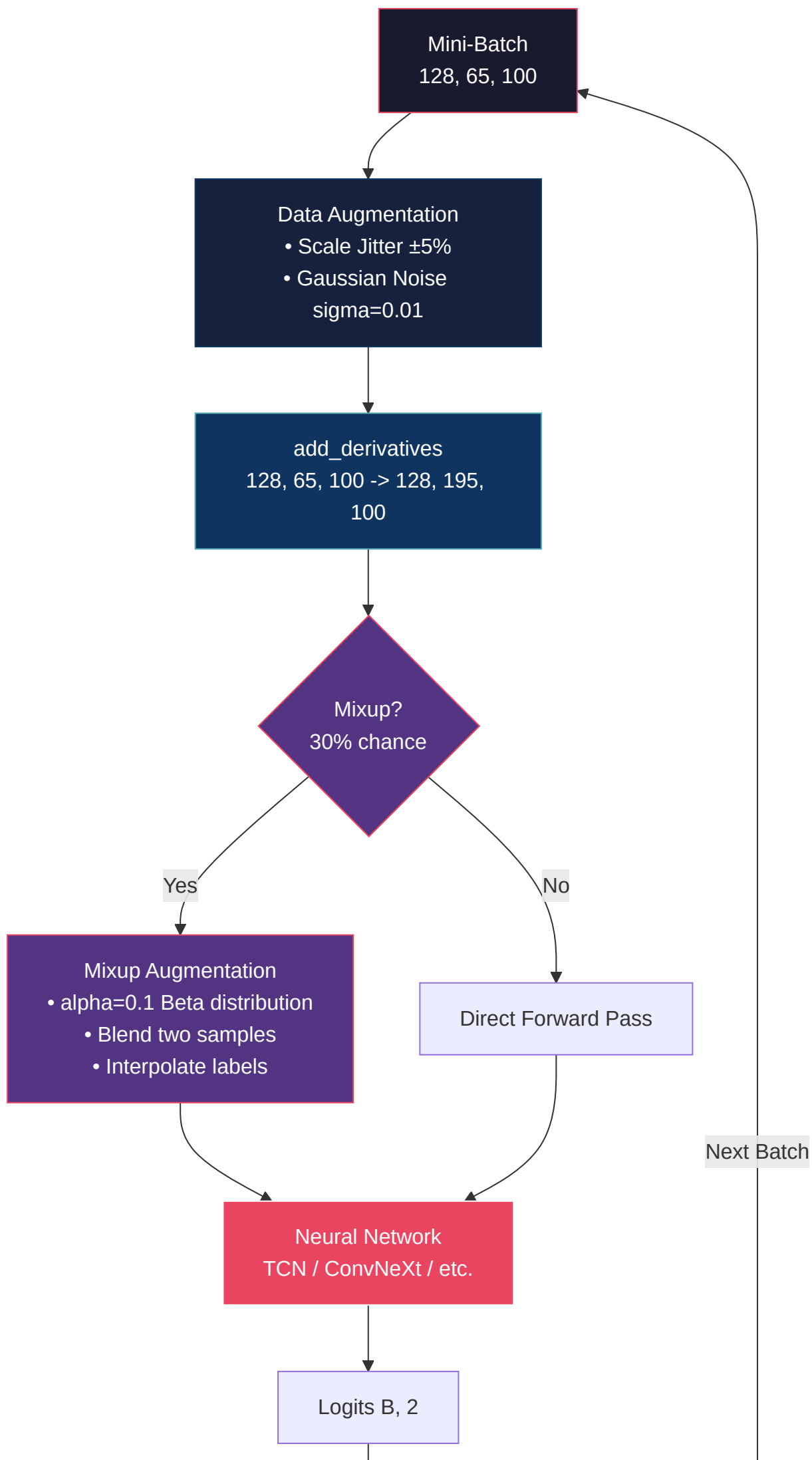
## 5. Cross-Validation Strategy

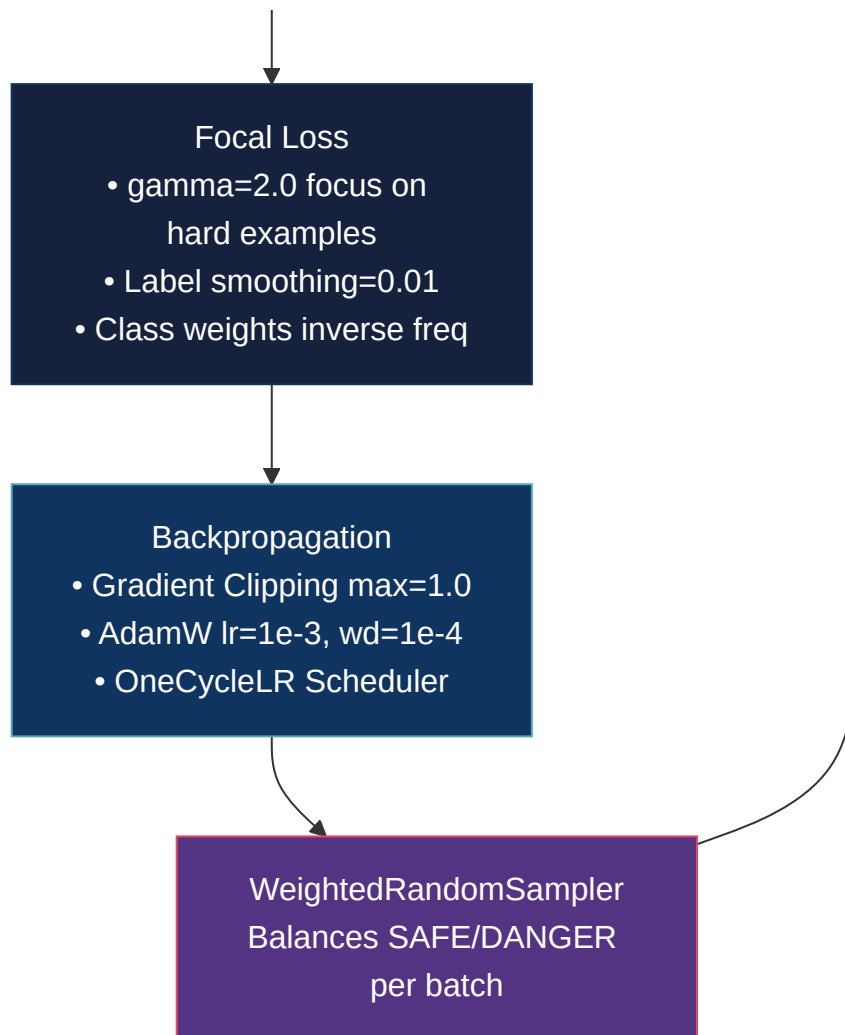


### 🔥 Important

**Subject-Dependent (Factory-Calibrated):** StratifiedKFold allows the model to see portions of every subject's data during training. This simulates a real factory where the cobot is calibrated to the specific workers on that floor.

## 6. Training Loop Architecture





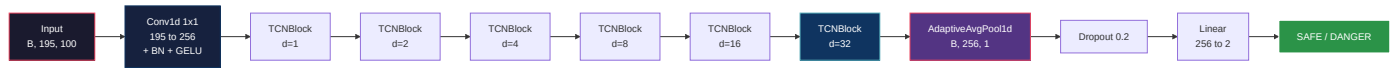
| Training Hyperparameter | Value                                  |
|-------------------------|----------------------------------------|
| Epochs                  | 30                                     |
| Batch Size              | 128                                    |
| Optimizer               | AdamW (lr=1e-3, weight_decay=1e-4)     |
| Scheduler               | OneCycleLR (max_lr=2e-3)               |
| Loss Function           | Focal Loss (gamma=2.0, smoothing=0.01) |
| Gradient Clipping       | max_norm=1.0                           |
| Mixup Probability       | 30%                                    |
| Mixup Alpha             | 0.1                                    |

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## 7. Model Architectures

### 7.1 TCN (Temporal Convolutional Network) — Best Overall



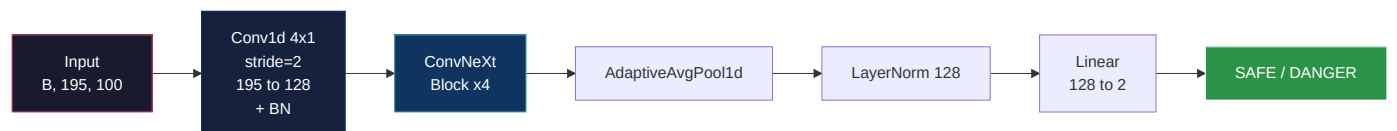


## TCNBlock (Residual):

```

Input ----+---- Conv1d(k=3, dilation=d) -> BN -> GELU -> Conv1d(k=3,
dilation=d) -> BN -> GELU ----+---- GELU -> Output
      |
      +-----+-----+-----+-----+-----+-----+-----+-----+-----+
      |                                                     Residual Skip -----+
      +-----+
  
```

## 7.2 ConvNeXt 1D

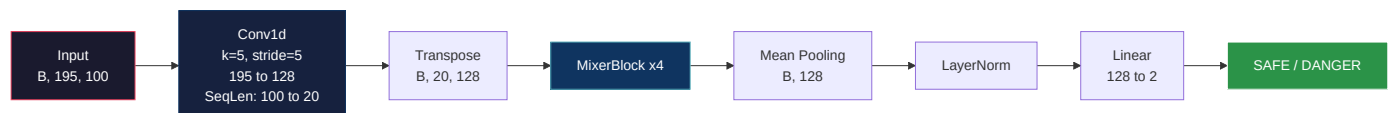


## ConvNeXtBlock (Inverted Bottleneck):

```

Input ----+---- DWConv1d(k=7, groups=dim) -> LayerNorm -> Linear(dim to 4*dim)
-> GELU -> Linear(4*dim to dim) ----+---- Output
      |
      +-----+-----+-----+-----+-----+-----+-----+-----+-----+
      |                                                     Residual Skip -----+
      +-----+
  
```

## 7.3 MLP-Mixer 1D



## MixerBlock:

```

Input ----+---- LayerNorm -> Transpose -> Linear(20 to 40) -> GELU ->
Linear(40 to 20) -> Transpose ----+---- (Token Mixing)
      |
      +-----+-----+-----+-----+-----+-----+-----+-----+-----+
      |                                                     Residual -----+
      +-----+
      ----+---- LayerNorm -> Linear(128 to 256) -> GELU -> Linear(256 to 128)
  
```

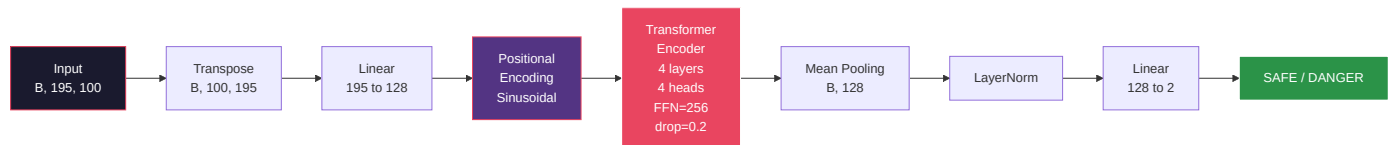
---+--- (Channel Mixing) -> Output

|

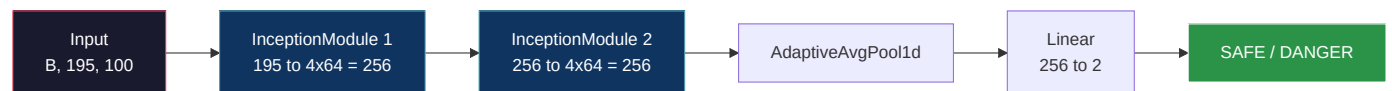
+----- Residual -----

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## 7.4 Transformer 1D



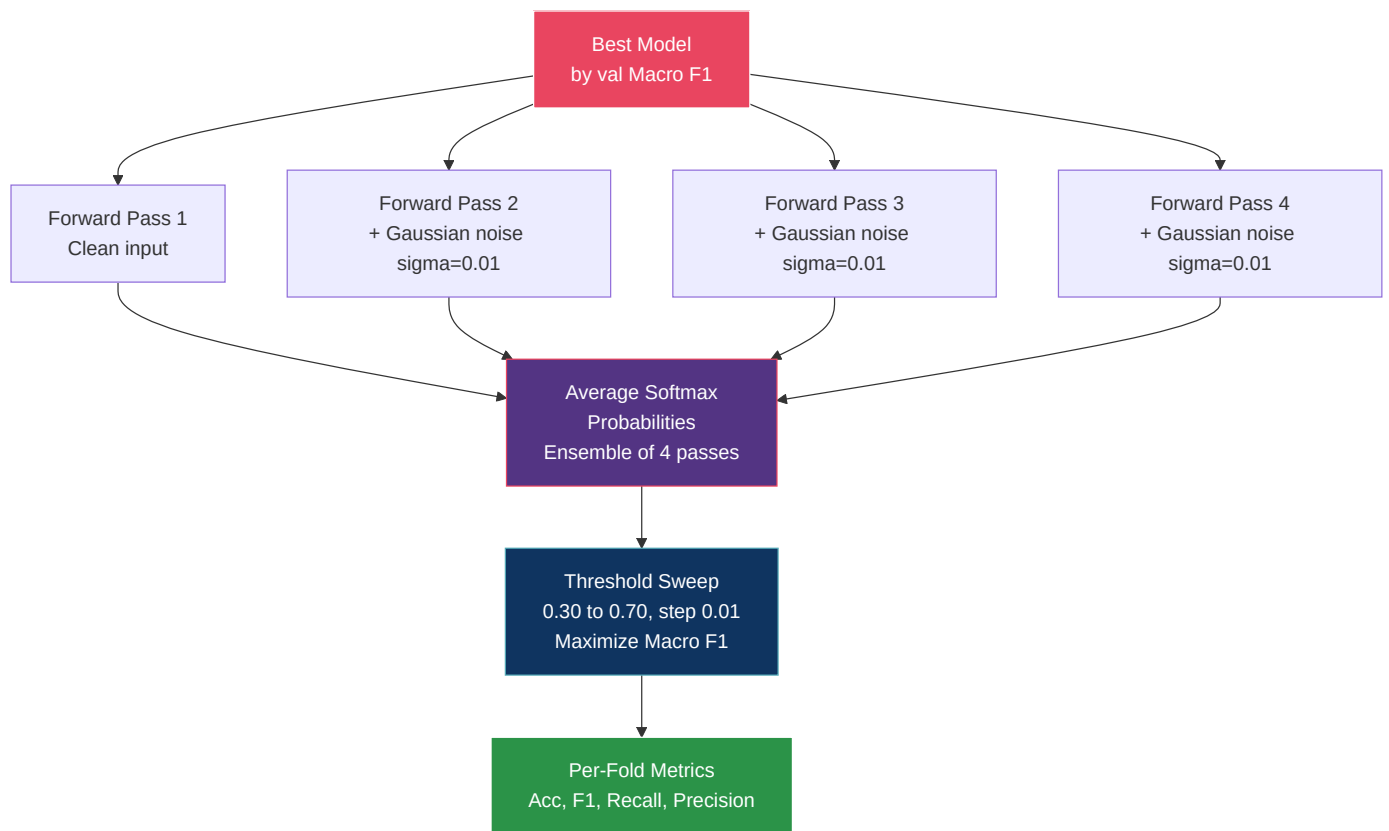
## 7.5 InceptionTime



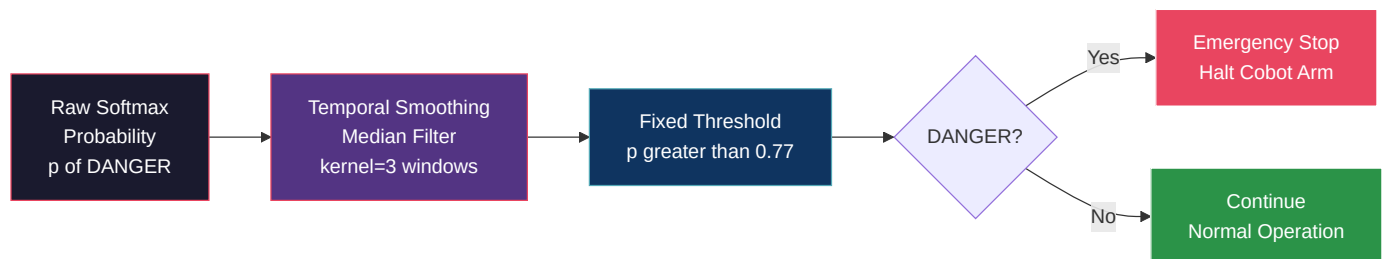
**InceptionModule (Multi-Scale):**

```
Input ---+--- Conv1d(k=1) -----+
      |                                     |
      +--- Conv1d(k=3, pad=1) -----+---
Concatenate -> BatchNorm -> GELU -> Output
      |                                     |
      +--- Conv1d(k=5, pad=2) -----+
      |                                     |
      +--- MaxPool1d(k=3, pad=1) -> Conv1d(k=1) -----+
```

## 8. Validation & Test-Time Augmentation (TTA)



## 9. Post-Processing Pipeline (Production Deployment)



### Impact of Post-Processing:

| Metric           | Raw Model | + Smoothing + Threshold 0.77 |
|------------------|-----------|------------------------------|
| Danger Precision | 83.92%    | <b>95.09%</b>                |
| Danger Recall    | 82.84%    | <b>98.44%</b>                |
| False Positives  | ~1,324    | <b>434 (-67%)</b>            |
| False Negatives  | ~142      | <b>133 (-6%)</b>             |

# 10. Complete Data Flow Summary

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### Phase 1: Data Pipeline

60 Subjects  
65 IMU Channels  
200 Hz

179 Trials  
Labelled: SAFE / DANGER

69094 Windows  
0.5s, 50% overlap  
Z-Score Normalised

### Phase 2: Training x5 Folds x 30 Epochs

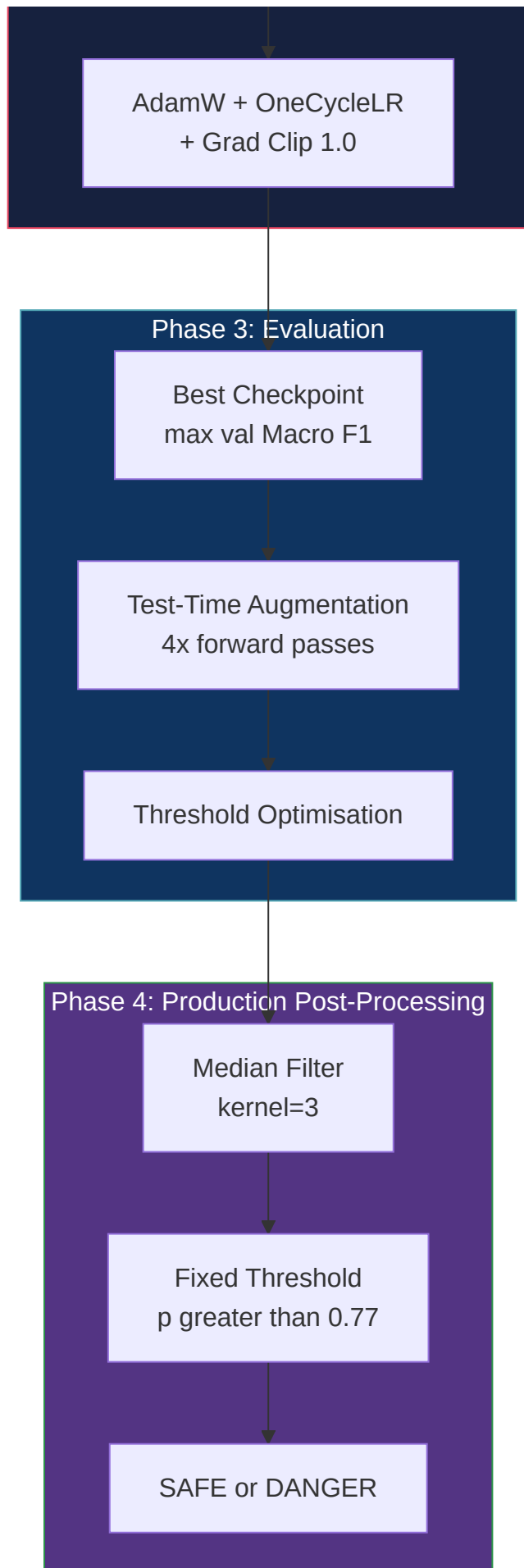
WeightedRandomSampler  
Balance SAFE/DANGER

GPU: add\_derivatives  
65 to 195 channels

Augment: Scale Jitter  
+ Noise + Mixup 30%

Model Forward Pass

Focal Loss  
gamma=2.0, weighted



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## 11. Final Model Rankings

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| Rank | Model          | Accuracy | Macro F1 | Danger Recall | Danger Precision |
|------|----------------|----------|----------|---------------|------------------|
| 1    | TCN            | 99.22%   | 97.88%   | 98.34%        | 95.09%           |
| 2    | ConvNeXt 1D    | 99.17%   | 97.76%   | 97.84%        | 95.62%           |
| 3    | MLP-Mixer 1D   | 99.28%   | 97.34%   | 95.87%        | 95.49%           |
| 4    | InceptionTime  | 98.30%   | 96.22%   | 97.53%        | 89.62%           |
| 5    | Transformer 1D | 96.47%   | 92.50%   | 95.85%        | 79.71%           |

 **Note**

Rankings are based on the **0.77 threshold + Median Filter** post-processing configuration. All metrics are computed over the full 69,094-window dataset using saved best weights from 5-fold cross-validation.